Related work matrix

Paper	Frameworks						Use cases						Metrics						Link rate	Same code
	XDP	AF_XDP	DPDK	Snabb	Linux stack	Other	Drop	L2FWD	L3FWD	Encap/Decap	SFC	Other	Throughput	Latency	Jitter	CPU	Memory/cache	Power		
Gallenmüller et al. (2015) [1]			✓			netmap, PF_RING ZC		✓				emulated CPU load, random memory access	✓	✓		✓	✓		10G	
Høiland-Jørgensen et al. (2018) [2]	✓		✓		✓	IPVS	✓	✓	✓	✓		DoS filter, L4 load balancing	✓	✓		✓			100G[c]	
Hohlfeld et al. (2019) [3]	✓	✓[a]				SmartNIC offload (XDP)	✓					emulated CPU load, DNS responder, map lookup/update	✓	✓		✓			10G	✓[b]

✓ measured empty = not covered

[a] Generic (copy) mode only, the NIC driver had no driver-specific AF_XDP mode on Linux 4.18.

[b] Same XDP program at all three execution points.

[c] At 64 B limited by PCIe: 115 Mpps drop, 70 Mpps forwarding out the same NIC.

References

[1] S. Gallenmüller, P. Emmerich, F. Wohlfart, D. Raumer and G. Carle. *Comparison of Frameworks for High-Performance Packet IO*. ANCS, 2015. https://ieeexplore.ieee.org/document/7110118

[2] T. Høiland-Jørgensen, J. D. Brouer, D. Borkmann, J. Fastabend, T. Herbert, D. Ahern and D. Miller. *The eXpress Data Path: Fast Programmable Packet Processing in the Operating System Kernel*. CoNEXT, 2018. https://dl.acm.org/doi/10.1145/3281411.3281443

[3] O. Hohlfeld, J. Krude, J. H. Reelfs, J. Rüth and K. Wehrle. *Demystifying the Performance of XDP BPF*. NetSoft, 2019. https://ieeexplore.ieee.org/document/8806651